



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester II Examinations 2024-2025

Course instance codes 3BA1, 4BMS2, 4BS2, 4BME1, 1EM1
Module Advanced Group Theory
Module code MA4344
Paper no. 1
External examiner Prof. Colva Roney-Dougal
Internal examiners Dr Niall Madden
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Instructions Answer all questions; show all working

Duration 2 hours
No. Pages 3 pages (including this cover page)
School Mathematical and Statistical Sciences
Discipline Mathematics

Requirements

Release in exam venue	Yes
Release to Library	Yes
Log tables	No
Other materials	No

1. (a) Define $x_m = \begin{pmatrix} 1+2m & 4m \\ -m & 1-2m \end{pmatrix}$. Let S be the set of all x_m as m ranges over $\mathbb{Z}$. Prove that S is a subgroup of $\text{SL}(2, \mathbb{Z})$ isomorphic to $\mathbb{Z}$. **10 marks**
- (b) (i) State (but do not prove) Lagrange's theorem and the fundamental isomorphism theorem for groups. **6 marks**
- (ii) Let G and H be finite groups with coprime orders. Prove that the only homomorphism between G and H is the trivial homomorphism. **9 marks**

2. (a) Let G be a group, and let $H \leq G$.
 - (i) Prove that the number $|G : H|$ of distinct left cosets of H in G is equal to the number of distinct right cosets of H in G . **5 marks**
 - (ii) Prove that if $|G : H| = 2$ then H is a normal subgroup of G . **7 marks**
- (b) What is a *simple group*? Prove that an abelian simple group is cyclic of prime order. **5 marks**
- (c) Let G be a group with centre $Z(G)$.
 - (i) Prove that if $G/Z(G)$ is a cyclic group then it is trivial. **6 marks**
 - (ii) Give an example of non-abelian G such that $G/Z(G)$ is abelian. **2 marks**

3. (a) Demonstrate an action of $C_2 \times C_2$ on the set of its elements, and use that action to construct an explicit isomorphism of $C_2 \times C_2$ onto a subgroup of $\text{Sym}(4)$. **9 marks**
- (b) Let G be a nontrivial finite group.
 - (i) Using the conjugation action of G on itself, derive the class equation: $|G| = |Z(G)| + \sum_{i=1}^t |G : C_G(y_i)|$ where $y_1, \dots, y_t$ are representatives of the distinct conjugacy classes of G not in $Z(G)$, and $C_G(y_i) = \{x \in G \mid xy_i = y_i x\}$. **6 marks**
 - (ii) Suppose that G has prime-power order. Use the class equation to prove that $Z(G)$ is nontrivial. **5 marks**
 - (iii) Let p be a prime. Using Question 2(c)(i) and part (b)(ii) of this question, prove that a group of order p^2 is abelian. **5 marks**

4. (a) Let G be a finite abelian group of order divisible by a prime p . Using induction on $|G|$, prove Cauchy's theorem, i.e., prove that G contains an element of order p . **7 marks**
- (b) State without proof the three parts (existence, uniqueness, arithmetic) of Sylow's Theorem. **6 marks**
- (c) Explain why there is no finite simple group of order
- (i) 30 **5 marks**
 - (ii) 36 **4 marks**
 - (iii) 2025. **3 marks**